Application Control (WDAC)

Implementation Solution Guide

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Executive Summary — Application Control (WDAC) (Microsoft Intune)

White-label · Intune Solutions

Antivirus asks "is this file known-bad?" **Application Control** asks the stronger question: "is this code explicitly trusted?" — and blocks everything else. **Windows Defender Application Control (WDAC / App Control for Business)** is kernel-enforced allowlisting that stops untrusted executables, scripts, and installers from running at all. Managed through Microsoft Intune and rolled out **Audit-first**, it delivers powerful protection without locking your users out.

This is a complete, **deploy-it-today** package for implementing WDAC with Microsoft Intune, written for an administrator with no prior application-control experience — including the audit-first method and recovery steps that make a powerful control safe to deploy.

What you get

- A plain-language **Solution Overview** of how allowlisting and trust models work.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click Intune steps in the required order.
- Five **PowerShell scripts** for policy discovery, validation, audit-event reporting, remediation/recovery, and tenant policy reporting.
- **Testing** and **Troubleshooting** guides for confident, lockout-free rollout.

What this delivers

An Intune **App Control for Business** policy that trusts **Windows + Microsoft Store** apps (and, optionally, reputable apps via the Intelligent Security Graph and **Intune-managed installers**), deployed in **Audit mode first**, reviewed against Code Integrity events, then moved to **Enforce**.

The standard deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Strong, kernel-enforced protection against untrusted code — deployed safely with audit-first validation and a clear recovery path — by a single administrator.

1. Solution Overview — Application Control (WDAC)

What WDAC is

Windows Defender Application Control (now **App Control for Business**) enforces an **allowlist**: only code that is explicitly trusted is allowed to run; everything else is blocked. It is enforced deep in Windows (kernel

level), making it far harder to bypass than user-mode controls. This solution manages it through Microsoft Intune.

Trust models (how code gets allowed)

- **Windows components + Microsoft Store apps** — trusted by the built-in base policy.
- **Reputable apps (Intelligent Security Graph / ISG)** — optionally trust apps Microsoft's cloud reputation considers good.
- **Intune-managed installers** — optionally trust anything installed by Intune (your deployed apps become trusted automatically).
- **Custom rules** — publisher, path, or file-hash rules via custom policy XML for advanced needs.

Modes

- **Audit only** — nothing is blocked; Windows **logs** what *would* have been blocked (Code Integrity events). This is how you roll out safely.
- **Enforced** — untrusted code is blocked.

> ***Method:*** *deploy in* ***Audit*** *first, review Code Integrity events, add the legitimate apps to trust, then switch to* ***Enforce****.*

How it works

1. An Intune **App Control for Business** policy sets the trust model and mode.
2. In **Audit**, the device logs **Code Integrity** events for code that isn't trusted.
3. You review those events, add legitimate apps to trust (managed installer / ISG / supplemental policy).
4. You switch the policy to **Enforce**; untrusted code is now blocked.

Benefits

- **Stops untrusted code** — including unknown malware, living-off-the-land binaries, and unsanctioned apps.
- **Kernel-enforced** — much harder to bypass than user-mode allowlisting.
- **Safe rollout** — Audit mode reveals impact before enforcement.
- **Low maintenance with managed installers** — apps you deploy via Intune are trusted automatically.

Limitations (be honest with buyers)

- WDAC is **powerful** — a misconfigured enforced policy can block legitimate apps. Audit-first + a recovery path are essential.
- Highly dynamic environments (developers compiling code, frequent new unsigned tools) need careful trust design.

• Custom XML policies are advanced; this solution favors the **managed-installer + ISG** model for most fleets.

• Recovery from an over-strict enforced policy is via removing/replacing the policy (covered here).

The standard deployment model

> ***Create the App Control policy in Audit → review Code Integrity events → add trust → Enforce → expand.*** *repeatable pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Application Control (WDAC)

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Licensing

Description: Intune (Plan 1). WDAC/App Control is a Windows feature (no separate license). **Why:** Intune delivers and reports the App Control policy. **Verify:** M365 admin center → **Billing** → **Licenses**. **Implement:** Assign M365 E3/E5 or Business Premium (Intune included).

2.2 Supported Windows OS

Description: Windows 10 1903+ or Windows 11 (App Control for Business CSP). **Why:** The Intune App Control policy and managed-installer trust require a supported build. **Verify:** `winver`; run `WDAC-Discovery.ps1`. **Implement:** Patch/upgrade to a supported build.

2.3 Device enrollment

Description: Devices enrolled in Intune. **Why:** The policy applies only to enrolled devices. **Verify:** Intune → **Devices** → **All devices**. **Implement:** Enable automatic enrollment; enroll devices.

2.4 Application inventory

Description: Awareness of the apps users legitimately run, and how they are installed. **Why:** Audit mode reveals untrusted apps; you must decide how each becomes trusted (managed installer, ISG, signature). **Verify:** Review `WDAC-AuditReport.ps1` output during the audit phase. **Implement:** Prefer deploying apps via **Intune (managed installer)** so they're trusted automatically; document any exceptions.

2.5 App deployment via Intune (recommended)

Description: Business apps deployed through Intune as managed installs. **Why:** With the managed-installer trust option, Intune-deployed apps are automatically allowed. **Verify:** Intune → **Apps** → **Windows**. **Implement:** Move key app deployments into Intune before enforcing.

2.6 Recovery plan

Description: A way to remove/replace the policy if enforcement is too strict. **Why:** WDAC is powerful; you need a safe rollback. **Verify:** Confirm `WDAC-Remediation.ps1` (policy-removal path) and an Intune unassign procedure. **Implement:** Document removing the assignment (and local policy clear) for recovery.

Checklist

- Intune licensing assigned
- Windows 10 1903+/11 — `WDAC-Discovery.ps1`
- Devices enrolled
- App inventory + trust approach decided (prefer managed installer)
- Key apps deployed via Intune
- Recovery/rollback procedure confirmed
- Pilot Entra group created

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:**

Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Application Control (WDAC)

Follow these steps **in order**, in the **Microsoft Intune admin center** (`https://intune.microsoft.com`).

> ***Model:*** *App Control for Business policy trusting Windows + Store (+ managed installer + ISG), deployed* ***Audit-first****, reviewed, then* ***Enforced****.*

Step 1 — Create the pilot group

1. Entra → **Groups** → **New group** → Security → `WDAC-Pilot` → add a few pilot devices. **Create**.

Step 2 — Create the App Control policy (Audit mode)

1. Intune → **Endpoint security** → **Application control** (App Control for Business) → **Create Policy**.

*(If not present in your tenant, use **Devices** → **Configuration** → **Create** → **Settings catalog** and search **"App Control" / "Application Control"**.)*

1. **Platform:** Windows. **Profile: App Control for Business**. **Create**.
2. **Name:** `WDAC - Audit (Pilot)`. **Next**.

Step 3 — Configure trust and mode

On **Configuration settings**:

1. **Application control code integrity policies** = **Enable**.
2. **Trust mode** / built-in policy = **Trust Windows components and Store apps**.
3. **Also trust apps with good reputation (ISG)** = **Enable** (recommended).
4. **Trust apps from managed installers** = **Enable** (so Intune-deployed apps are allowed).
5. **Enforcement** = **Audit only** for now.

Click **Next**.

Step 4 — Assign and run the audit

1. **Assignments** → **Add groups** → `WDAC-Pilot`. **Next** → **Create**.
2. Let pilot devices run normally for **2+ weeks**, exercising all line-of-business apps.

Step 5 — Review Code Integrity audit events

1. On pilot devices run `WDAC-AuditReport.ps1` (reads Code Integrity audit events 3076).
2. Identify legitimate apps that *would* have been blocked.

Step 6 — Add trust for legitimate apps

1. Preferred: **deploy those apps via Intune** so the **managed installer** trust covers them.
2. For signed apps not from Intune, rely on **ISG** reputation or add a **supplemental/custom policy** (advanced).
3. Re-audit until the audit log is clean of legitimate blocks.

Step 7 — Switch to Enforce

1. Duplicate the policy as `WDAC - Enforce (Pilot)` (or edit) and set **Enforcement = Enforced**. **Save**.
2. Re-validate with `WDAC-Validation.ps1` (confirms an enforced policy is active).
3. Confirm pilot users can run all required apps; untrusted code is blocked.

Step 8 — Expand to production

1. Add a production group to the **Enforce** policy **Assignments**; roll out in waves.
2. Keep new apps flowing through Intune (managed installer) so they stay trusted.
3. Re-validate each wave with `WDAC-Validation.ps1` / `WDAC-AuditReport.ps1`.

> ***Recovery:*** *if an enforced policy is too strict,* ***unassign*** *it in Intune (and run* `WDAC-Remediation.ps1` *to clear/refresh the local policy), then return to Audit, fix trust, re-enforce.*

Order of operations

`Pilot group → App Control policy (Audit) → Trust Windows/Store/ISG/managed-installer → Assign → Review audit → Add trust → Enforce → Expand`

5. Testing Guide — Application Control (WDAC)

Run during the audit phase (Implementation Step 5) and after switching to Enforce.

Test 1 — Policy applied

Do: Intune → **Endpoint security** → **Application control** → policy → **Device status**. **Expected:** Pilot devices show **Succeeded**.

Test 2 — Policy active on device

Do: On a pilot device run `WDAC-Validation.ps1` (checks for an active App Control policy and its mode). **Expected:** An App Control policy is active; mode reflects Audit (pilot) or Enforced (post-switch).

Test 3 — Audit events generated

Do: Run an untrusted test binary (a benign unsigned EXE) on an Audit-mode device, then run `WDAC-AuditReport.ps1`. **Expected:** A Code Integrity **audit** event (3076) is logged for the binary — proving the policy evaluates.

Test 4 — Enforce blocks untrusted code

Do: With the policy **Enforced**, run the same untrusted test binary. **Expected:** Execution is **blocked** (Code Integrity event 3077); the binary does not run.

Test 5 — Trusted apps still run

Do: With Enforce on, launch all line-of-business apps (deployed via Intune / Store / signed). **Expected:** All legitimate apps run normally; no false blocks.

Test 6 — Managed-installer trust works

Do: Deploy a new app via Intune to an enforced device and run it. **Expected:** The Intune-deployed app runs (trusted via managed installer) without manual rules.

Test 7 — Recovery path works

Do: On a test device, unassign the policy in Intune and run `WDAC-Remediation.ps1`. **Expected:** The enforced policy is removed/cleared and the device returns to normal — confirming rollback.

Pass/fail gate

Gate	Requirement
Audit pass	Tests 1–3 pass; audit log clean of legitimate-app blocks after trust added
Enforce pass	Tests 4–6 pass (no false blocks); Test 7 recovery confirmed

6. Troubleshooting Guide — Application Control (WDAC)

Use `WDAC-Validation.ps1` / `WDAC-Discovery.ps1` for policy state and `WDAC-AuditReport.ps1` for events, `WDAC-Remediation.ps1` for recovery.

A. Legitimate app blocked (enforced)

Symptom	Cause	Remediation
A required app won't run after Enforce	It isn't trusted (not Intune-deployed, not ISG-reputable, unsigned)	Identify via `WDAC-AuditReport.ps1`; deploy it via Intune (managed installer), or add a supplemental/custom rule; if urgent, unassign the policy to recover
Whole app category blocked	Trust model too narrow	Enable **ISG** and **managed installer** trust; re-audit before re-enforcing

B. Recovery from an over-strict policy

1. **Unassign** the App Control policy from the device group in Intune.
2. Run `WDAC-Remediation.ps1` to clear/refresh the local policy and force a sync.
3. The device returns to its prior state; return the policy to **Audit**, fix trust, then re-enforce.

WDAC policies persist until replaced/removed and may require a reboot to fully clear.

C. Policy not applying

Symptom	Cause	Remediation
No policy active on device	Not synced / not enrolled / unsupported OS	`WDAC-Remediation.ps1` (sync); confirm Windows 1903+/11 and group membership
Audit shows nothing	Users didn't run untrusted code, or wrong log	Run a benign unsigned test EXE; check the Code Integrity Operational log

D. Too many audit events

• Expected initially — every not-yet-trusted app logs. Triage by publisher/path with `WDAC-AuditReport.ps1`; move the common legitimate ones to managed-installer deployment first.

E. Diagnostics quick reference

```
# active policies (Windows 11):
citool.exe --list-policies
# Device Guard / code integrity status:
Get-CimInstance -Namespace root\Microsoft\Windows\DeviceGuard -ClassName Win32_DeviceGuard | Select CodeIn
# Code Integrity events: 3076 (audit block), 3077 (enforced block), 3099 (policy loaded)
Get-WinEvent -LogName 'Microsoft-Windows-CodeIntegrity/Operational' | Where-Object Id -in 3076,3077,3099
```

F. When to escalate

If a signed, Intune-deployed app is still blocked under Enforce with managed-installer trust enabled, capture the 3077 event detail (file path, signing info) and open a Microsoft support case.

7. Best Practices — Application Control (WDAC)

• **Audit first, always.** WDAC is powerful — never deploy Enforced without a thorough audit phase. This is the #1 rule.

• **Have a recovery path ready.** Know how to unassign the policy and clear it locally before you enforce anything.

• **Prefer the managed-installer model.** Deploy apps via Intune so they're trusted automatically — far lower maintenance than custom XML rules.

• **Enable ISG reputation.** It trusts well-known good apps without per-app rules, reducing audit noise.

• **Pilot on representative devices.** Include devices that run your full app set so the audit reveals real blocks.

• **Triage audit events by publisher/path.** Trust whole trusted publishers rather than individual hashes where possible.

• **Keep new apps flowing through Intune.** That keeps them trusted automatically as your estate changes.

• **Avoid custom XML unless needed.** Reserve custom policies for genuine edge cases; they add maintenance burden.

• **Communicate to users.** Let pilot users report anything that "won't open" during audit so you catch edge cases.

• **Document the trust model.** Record what's trusted (Windows/Store/ISG/managed installer/custom) and the recovery procedure for audits.

8. Deliverables Summary — Application Control (WDAC)

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What WDAC is, trust models, modes, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps — Audit-first, add trust, Enforce, with recovery
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and recovery from over-strict policies
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`WDAC-Discovery.ps1`	Reports OS support and current code-integrity/App Control status
`WDAC-Validation.ps1`	Confirms an App Control policy is active and its enforcement mode
`WDAC-AuditReport.ps1`	Parses Code Integrity events (3076/3077) to find blocked/would-block code
`WDAC-Remediation.ps1`	Recovery: clears/refreshes local policy and forces a sync
`WDAC-PolicyReport.ps1`	Tenant-wide App Control policy assignment report (Microsoft Graph)

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Strong, kernel-enforced protection against untrusted code — deployed safely with audit-first validation and a clear recovery path.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

WDAC-AuditReport.ps1

WDAC-Discovery.ps1

WDAC-PolicyReport.ps1

WDAC-Remediation.ps1

WDAC-Validation.ps1